

Performance Testing

Date	30 June 2026
Team ID	
Project name	Opticrop
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter, Flask Testing
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Crop Prediction Module
Test Environment	Local System / Cloud Environment
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Multiple users accessing crop prediction page	20	60	System should respond without failure
2	Simultaneous login requests	15	45	Authentication should work properly
3	Large input data testing	10	30	System should process data correctly
4	Database connectivity under load	12	40	Database should store and retrieve data without delay

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 seconds	Pass	Response time within limit
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Stable under load
3	Throughput (Req/sec)	> 20	25	Pass	Good request handling
4	Error Rate	< 1%	0.2%	Pass	Minimal errors observed
5	CPU Utilization	< 80%	65%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	70%	Pass	Memory usage acceptable

Step 4: Observations & Analysis**Key Findings:**

The OptiCrop application performed efficiently under expected user load conditions. Response times remained stable and system performance met the defined benchmarks.

Bottlenecks Identified:

Minor delays were observed during simultaneous prediction requests with high input volume.

Optimization Steps Taken:

Input validation and backend request handling were optimized to improve system responsiveness and reduce processing delays.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

The image shows two screenshots of the OptiCrop website. The top screenshot is the homepage, featuring a green header with the 'OptiCrop' logo and navigation links (Home, About, FindYourCrop). The main content area has a background image of a field at sunset with the text 'AI + MACHINE LEARNING FOR AGRICULTURE' and 'Smart crop recommendations from soil and climate values'. Below this is a green button labeled 'Find Your Crop'. The bottom section of the homepage highlights three features: 'Data-driven' (uses crop recommendation data to connect field conditions with crop labels), 'Practical' (simple web form for quick field-condition entry and prediction), and 'Sustainable' (supports nutrient-aware, rainfall-aware, and resource-conscious planning).

The bottom screenshot shows the 'Find Your Crop' form. It has a title 'Find Your Crop' and a subtitle 'Enter soil nutrient and environmental values to get a crop recommendation.' The form contains several input fields for different parameters:

Parameter	Value
Nitrogen (N)	90
Phosphorous (P)	42
Potassium (K)	43
Temperature	20.8
Humidity	82
pH	6.5
Rainfall	202

At the bottom of the form is a green button labeled 'Predict Crop'. Below the form, the text 'Smart Agricultural Production Optimization Engine' is displayed.